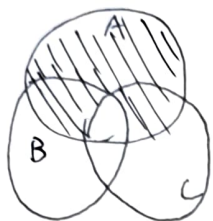


(a)



(b)



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HW2

2. Let f 's domain is B , co-domain is C
 g 's domain is A , co-domain is B

(a)

f is onto $= \forall y \in C \exists x \in B$ s.t. $f(x) = y$

$f \circ g$ is onto $: \forall a \in B \exists x \in A$ s.t. $g(x) = a$ — ①
 and

$f(g(x))$

$\forall y \in C \exists a \in B$ s.t. $f(a) = y$ — ②

Det: if g is onto $= \forall y \in B \exists x \in A$ s.t. $g(x) = a$

from ①

if f and $f \circ g$ is onto $\Rightarrow g$ is onto.

(b)

f is 1-1 $: f(a) = f(b) \Leftrightarrow a = b$ — ①

$f \circ g$ is 1-1 $: f(g(a)) = f(g(b)) \Leftrightarrow a = b$ — ②

Det: if g is 1-1 $: g(a) = g(b) \Leftrightarrow a = b$

$f(g(a)) = f(g(b)) \Leftrightarrow g(a) = g(b)$

from ①

$\Leftrightarrow a = b$

from ②

$\therefore g(a) = g(b) \Leftrightarrow a = b$

$\Rightarrow g$ is 1-1.

3.

(a)

$$a_0 = 2$$

$$a_1 = 6 \cdot a_0 = 12$$

$$a_2 = 6 \cdot a_1 = 72$$

$$a_3 = 6 \cdot a_2 = 432$$

$$a_4 = 6 \cdot a_3 = 2592$$

(b)

$$a_0 = 1$$

$$a_1 = 1$$

$$a_2 = 2 \cdot a_1 + 2^2 \cdot a_0 = 2 \cdot 1 + 4 \cdot 1 = 6$$

$$a_3 = 3 \cdot a_2 + 3^2 \cdot a_1 = 3 \cdot 6 + 9 \cdot 1 = 27$$

$$a_4 = 4 \cdot a_3 + 4^2 \cdot a_2 = 4 \cdot 27 + 16 \cdot 6 = 204$$

(c)

$$a_0 = 1$$

$$a_1 = 2$$

$$a_2 = 0$$

$$a_3 = a_2 + a_0 = 1$$

$$a_4 = a_3 + a_1 = 3$$

(d)

$$a_0 = -1$$

$$a_1 = 0$$

$$a_2 = 2 \cdot a_1 + a_0^2 = 2 \cdot 0 + 1 = 1$$

$$a_3 = 3 \cdot a_2 + a_1^2 = 3 \cdot 1 + 0 = 3$$

$$a_4 = 4 \cdot a_3 + a_2^2 = 4 \cdot 3 + 1 = 13$$

(e)

$$a_0 = 1$$

$$a_1 = 1$$

$$a_2 = 2$$

$$a_3 = a_2 - a_1 + a_0 = 2 - 1 + 1 = 2$$

$$a_4 = a_3 - a_2 + a_1 = 2 - 2 + 1 = 1$$

4.

(a)

$$a_0 = -1$$

$$a_1 = 2a_0 - 3 = -5 \quad \downarrow -4$$

$$a_2 = 2a_1 - 3 = -13 \quad \downarrow -8$$

$$a_3 = 2a_2 - 3 = -29 \quad \downarrow -16$$

$$a_3 = 2a_2 - 3 = -29$$

$$\vdots$$

$$a_n = 2a_{n-1} - 3$$

$$\Rightarrow$$

$$a_0 = -1$$

$$a_1 = a_0 - 2^2$$

$$a_2 = a_1 - 2^3$$

$$a_3 = a_2 - 2^4$$

$$\vdots$$

$$+ | a_n = a_{n-1} - 2^{n+1}$$

$$a_n = -1 - 2^2 - 2^3 - 2^4 - \dots - 2^{n+1}$$

$$= -(2^1 + 2^2 + \dots + 2^{n+1}) + 1$$

$$= - \frac{2(2^{n+1} - 1)}{2 - 1} + 1$$

$$= -2^{n+2} + 2 + 1$$

$$a_n = -2^{n+2} + 3$$

4.

$$(b) \quad a_0 = 4$$

$$a_1 = a_0 + 2 \cdot 1 + 3 = 9$$

$$a_2 = a_1 + 2 \cdot 2 + 3 =$$

$$\vdots$$

$$\text{+ } \underline{a_n = a_{n-1} + 2n + 3}$$

$$(a_0 + \dots + a_n) = 4 + (a_0 + \dots + a_{n-1}) + 2 \cdot (1 + 2 + \dots + n) + 3 \cdot n$$

$$a_n = 4 + 2 \cdot \frac{n(n-1)}{2} + 3n = 4 + n^2 - n + 3n$$

$$a_n = n^2 + 2n + 4$$

(c)

$$a_0 = 1$$

$$a_1 = 2 \cdot 1 \cdot a_0$$

$$a_2 = 2 \cdot 2 \cdot a_1$$

$$\vdots$$

$$\text{x } \underline{a_n = 2 \cdot n \cdot a_{n-1}}$$

$$(a_0 \times a_1 \times \dots \times a_n) = (a_0 \times a_1 \times \dots \times a_{n-1}) \times 2^n \times n!$$

$$a_n = 2^n \times n!$$

5. (a)

$$a_n = a_{n-1} + 2a_{n-2} + 2n - 9 = -n + 2$$

$$a_{n-1} = -(n-1) + 2 = -n + 3 \quad \text{--- ①}$$

$$a_{n-2} = -(n-2) + 2 = -n + 4 \quad \text{--- ②}$$

$$a_n = -n + 3 + 2(-n + 4) + 2n - 9 \quad \text{By ①, ②}$$

$$= -n + 3 + 2n + 8 - 2n - 9$$

$$= -n + 2$$

QED.

(b)

$$a_n = 3(-1)^n + 2^n - n + 2$$

$$\begin{aligned} a_{n-1} &= 3(-1)^{n-1} + 2^{n-1} - (n-1) + 2 \\ &= 3 \frac{(-1)^n}{-1} + \frac{2^n}{2} - n + 3 \end{aligned}$$

$$= -3(-1)^n + \frac{2^n}{2} - n + 3$$

$$a_{n-2} = 3 \cdot (-1)^{n-2} + 2^{n-2} - (n-2) + 2$$

$$= 3 \cdot (-1)^n + \frac{2^n}{4} - n + 4$$

$$a_n = a_{n-1} + 2a_{n-2} + 2n - 9$$

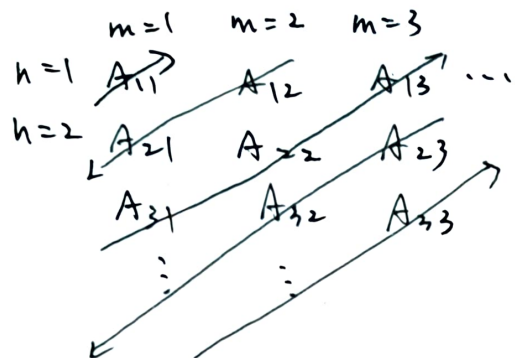
$$\begin{aligned} &= \underline{-3 \cdot (-1)^n + \frac{2^n}{2} - n + 3} + 2 \left[\underline{3 \cdot (-1)^n + \frac{2^n}{4} - n + 4} \right] \\ &\quad + 2n - 9 \end{aligned}$$

$$= 3(-1)^n + 2^n - n + 2 \quad \text{QED.}$$

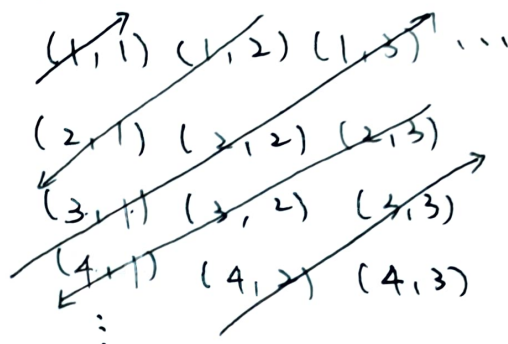
6.

(a)

Let A_{nm} is the m^{th} element of n^{th} countable set



(b)



7.

$$AB = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & -2 & -1 \\ 1 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 \cdot 3 + (-1) \cdot 1 & -2 & -1 \cdot 2 \\ 0 & 0 & 2 \\ 6 + 3 & -4 & -2 + 6 \end{bmatrix} = \begin{bmatrix} 2 & -2 & -3 \\ 0 & 0 & 2 \\ 9 & -4 & 4 \end{bmatrix} = (AB)_{33}$$

$A_{12} \quad B_{23}$

8.

$$\exists \text{ const } c, k, \quad |x^4 + 9x^3 + 4x + 7| \leq c|x^4|, \quad \forall x > k$$

$$|x^4 + 9x^3 + 4x + 7| \stackrel{x>1}{=} x^4 + 9x^3 + 4x + 7 \leq x^4 + 9x^4 + 4x^4 + 7x^4 \stackrel{c=21}{=} cx^4 = dx^4$$

$$\text{Let } c=21, k=1$$

$$|x^4 + 9x^3 + 4x + 7| \leq 21|x^4|, \quad \forall x > 1$$

$$\Rightarrow x^4 + 9x^3 + 4x + 7 \text{ is } O(x^4)$$

9.

$$\exists \text{ 常数 } c, k, \quad \left| \frac{(x^2+1)}{x+1} \right| \leq c|x|, \quad \forall x > k$$

$$|x^2+1| = x^2+1 \leq x^2+x^2 = 2x^2$$

$$|x+1| \stackrel{x>0}{=} x+1 \geq x$$

$$\left| \frac{x^2+1}{x+1} \right| \stackrel{x>0}{=} \frac{x^2+1}{x+1} \leq \frac{2x^2}{x} = 2x \stackrel{x>0}{=} 2|x|$$

$$\text{令 } c=2, k=0$$

$$\Rightarrow \frac{x^2+1}{x+1} \text{ 是 } O(x)$$

10.

By definition

$$\exists \text{ 常数 } c, k, \quad \left| \frac{x^3 + 2x}{2x+1} \right| \leq c |x^2|, \quad \forall x > k$$

$$|x^3 + 2x| \stackrel{x > 0}{=} x^3 + 2x \leq x^3 + 2x^3 \leq 3x^3$$

$$|2x+1| \stackrel{x > 0}{=} 2x+1 \geq 2x$$

$$\left| \frac{x^3 + 2x}{2x+1} \right| \stackrel{x > 0}{=} \frac{x^3 + 2x}{2x+1} \leq \frac{3x^3}{2x} = \frac{3}{2} x^2 = \frac{3}{2} |x^2|$$

$$\text{令 } c = \frac{3}{2}, k = 0$$

$$\left| \frac{x^3 + 2x}{2x+1} \right| \leq \frac{3}{2} |x^2|, \quad \forall x > 0$$

$$\Rightarrow \frac{x^3 + 2x}{2x+1} \text{ is } o(x^2)$$